ANISH SHIRODKAR

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SUMMARY

M.S. CS candidate at Rutgers University specializing in ML, NLP, and AI engineering. Builds and ships end-to-end AI-powered systems across computer vision (YOLOv8, ByteTrack), RAG-based NLP pipelines (ChromaDB, LLaMA 3.1, HuggingFace), and full-stack LLM applications (Gemini API, Supabase), from model to production. Published patent. Seeking Summer 2026 ML / AI / NLP Engineer internship.

EDUCATION

Rutgers University | M.S. Computer Science | GPA: 3.50 | Sept 2025 – Present

Thakur College of Engineering & Technology, University of Mumbai | B.Tech IoT | GPA: 3.67 | Dec 2021 – May 2025

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, SQL
ML / AI / NLP: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, LSTM, Attention Mechanisms, Transformers, Fine-tuning, PPO, DQN, RAG
LLM & Embeddings: Gemini API, Groq/LLaMA 3.1, HuggingFace Transformers, Sentence Transformers, ChromaDB, Vector Databases
Computer Vision: YOLOv8, ByteTrack, OpenCV, Ultralytics, ONNX, object detection, multi-object tracking, speed estimation
Web & Backend: React, Next.js, FastAPI, Flask, Streamlit, REST APIs, Supabase (PostgreSQL + RLS), Firebase, shadcn/ui
MLOps & Infra: Model Deployment, Inference, Docker, Hugging Face Spaces (GPU), Vercel, AWS, Weights & Biases, Git, CI/CD

PROJECTS

Vtrack: Real-Time Traffic Analysis System | 2026
YOLOv8s | ByteTrack | FastAPI | React 18 | React Three Fiber | Supabase | Docker | Hugging Face Spaces | Vercel

- Engineered end-to-end computer vision pipeline using YOLOv8s for real-time multi-class object detection (cars, trucks, buses, motorcycles) and ByteTrack for multi-object tracking with vehicle re-identification, achieving ~25-30 FPS throughput and ~90-95% vehicle count accuracy across test videos
- Deployed containerized FastAPI inference backend on Hugging Face Spaces (GPU) via Docker; built immersive 3D dashboard using React Three Fiber with live counters, directional counting lines, speed estimation via camera calibration, heatmap generation, and Supabase real-time data sync

SkillGap: AI Career Assistant | 2026
Next.js 15 | React 19 | Supabase | Gemini 2.5 Flash | pdf.js | shadcn/ui | Tailwind CSS

- Built a two-stage NLP pipeline using Gemini 2.5 Flash: extracts structured skill entities from raw PDF resume text, independently extracts skills from a target job description, then compares both sets to generate a ranked skill gap report with recommendations, validated across 15+ resumes
- Implemented a stateful multi-turn conversational AI interview simulator using Gemini 2.5 Flash; maintains conversation history across turns, generates role-specific technical questions grounded in the candidate's resume, and produces a structured performance evaluation with final grade at session end

DocPilot: NLP Documentation Assistant (RAG) | 2025
Python | RAG | ChromaDB | LLaMA 3.1 via Groq | HuggingFace Sentence Transformers | Streamlit

- Implemented a Retrieval-Augmented Generation (RAG) pipeline for NLP question answering over ~500 pages of HuggingFace Transformers, PyTorch, and Scikit-learn documentation; generated dense embeddings with HuggingFace sentence-transformers, indexed into a local ChromaDB vector database for semantic retrieval, and integrated LLaMA 3.1 via Groq for low-latency, source-cited, hallucination-reduced inference

Weather Forecasting Decision Support System | April 2025
Attention-LSTM | PyTorch | NumPy | Pandas | Flask | Firebase | AWS | NodeMCU

- Designed and trained an attention-based LSTM achieving ~85% prediction accuracy on multi-variable rainfall forecasting (humidity, pressure, moisture) from live IoT sensor streams; implemented full pipeline from NodeMCU hardware through Flask API to Firebase and AWS for real-time cloud inference (published as IP India patent, 2025)

EXPERIENCE

Indian Meteorological Department, Gujarat, ML Project Member | April 2023 – June 2025
Gujarat, India

- Led end-to-end ML model development and IoT-to-cloud data pipeline for a live government weather forecasting system; attention-LSTM architecture recognized as a government-registered patent (IP India, 2025)

CognoRise InfoTech, Artificial Intelligence Intern | April 2024 – June 2024
Mumbai, India

- Developed and optimized ML models for production; built modular Python data preprocessing and feature engineering pipelines using NumPy and Pandas to accelerate model training workflows

CERTIFICATIONS & PUBLICATIONS

Hugging Face: AI Agents Fundamentals | Mar 2026
Anthropic: Model Context Protocol: Advanced Topics | Mar 2026
Anthropic: Introduction to Agent Skills | Mar 2026
Shirodkar, A. V. *Smart Weather Forecasting: IoT-Integrated Decision Support System for Real-Time Analysis.* Intellectual Property India, 2025.